FV 1

OSNOVE MATEMATIKE

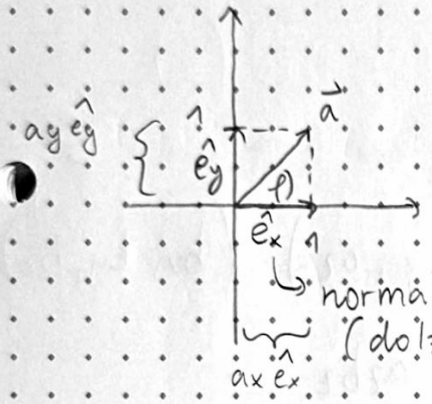
1. VEKTORJI

2D → urejeni pari števil

enotski vektor

$$\vec{a} = (a_x, a_y) = a_x \hat{e}_x + a_y \hat{e}_y$$

$\hat{i}$ $\hat{j}$ $\hat{k}$

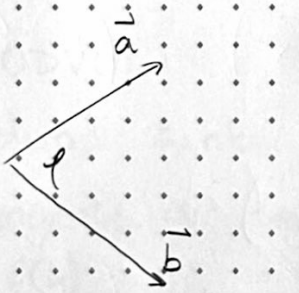


$$a_x = |\vec{a}| \cdot \cos\varphi$$

$$a_y = |\vec{a}| \cdot \sin\varphi$$

dolžina vektorja $|\vec{a}| = \sqrt{a_x^2 + a_y^2}$

skalarni produkt



$$\vec{a} \cdot \vec{b} = |\vec{a}| \cdot |\vec{b}| \cdot \cos\varphi$$

$$\vec{a} \cdot \vec{b} = (a_x, a_y) \cdot (b_x, b_y)$$

$$= a_x \cdot b_x + a_y \cdot b_y$$

2 vektorja računamo po komponentah

vsota: $\vec{a} + \vec{b} = (a_x, b_x) + (a_y, b_y)$

$$= (a_x + b_x, a_y + b_y)$$

množenje s skalarjem: $\alpha \vec{a} = \alpha (a_x, a_y) = (\alpha a_x, \alpha a_y)$

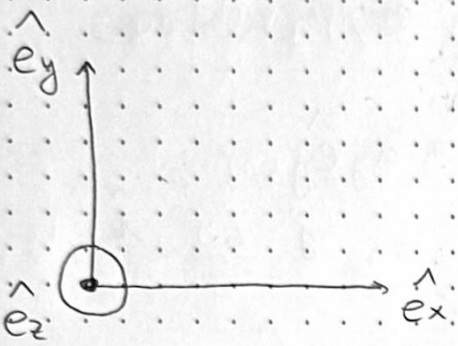
3D vektorji → urejene trojice števil

$\vec{a} = (a_x, a_y, a_z)$

$= a_x \hat{e}_x + a_y \hat{e}_y + a_z \hat{e}_z$

not v tablo

iz table



Skalarni produkt: $\vec{a} \cdot \vec{b} = (a_x, a_y, a_z)(b_x, b_y, b_z)$

(rešitev številka)

$= a_x b_x + a_y b_y + a_z b_z$

$= |\vec{a}| \cdot |\vec{b}| \cdot \cos\varphi$

Vsota:

$\vec{a} + \vec{b} = (a_x + b_x, a_y + b_y, a_z + b_z)$

$\alpha \vec{a} = (\alpha a_x, \alpha a_y, \alpha a_z)$

Vektorski produkt (vrstni red pomemben)

$\vec{c} = \vec{a} \times \vec{b} = \begin{vmatrix} \hat{e}_x & \hat{e}_y & \hat{e}_z \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix} \begin{matrix} \hat{e}_x & \hat{e}_y & \hat{e}_z \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{matrix}$

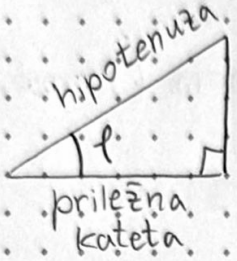
6 členov $= \hat{e}_x \cdot a_y \cdot b_z + \hat{e}_y \cdot a_z \cdot b_x + \hat{e}_z \cdot a_x \cdot b_y$

$- \hat{e}_x \cdot a_z \cdot b_y - \hat{e}_y \cdot a_x \cdot b_z - e_z \cdot a_y \cdot b_x$

$= \hat{e}_x (a_y \cdot b_z - a_z b_y) + \hat{e}_y (a_z b_x - a_x b_z) + \hat{e}_z (a_x b_y - a_y b_x)$

$(a_y b_z - a_z b_y, a_z b_x - a_x b_z, a_x b_y - a_y b_x)$

2. KOTNE FUNKCIJE



$$\sin\varphi = \frac{\text{nasprotna}}{\text{hipotenuza}}$$

$$\cos\varphi = \frac{\text{priležna}}{\text{hipotenuza}}$$

$$\tan\varphi = \frac{\text{nasprotna}}{\text{priležna}} = \frac{\sin\varphi}{\cos\varphi} = tg.\,\varphi$$

$$\cot\varphi = \frac{1}{\tan\varphi} = \frac{\text{priležna}}{\text{nasprotna}} = ctg.\,\varphi$$

$$\sin^2\varphi + \cos^2\varphi = 1$$

$$\sin\varphi = \frac{\tan\varphi}{\sqrt{1+\tan^2\varphi}} \qquad \cos\varphi = \frac{1}{\sqrt{1+\tan^2\varphi}}$$

IZRELJI!

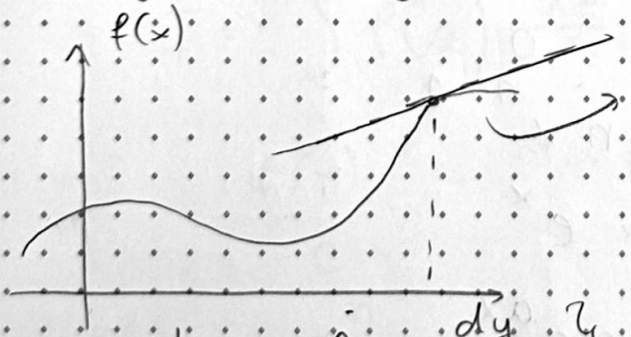
ADICIJSKI IZREKI ...

DVOJNI KOTI ...

FAKTORIZACIJE ...

3. ODVODI

Odvod funkcije f v točki x nam pove naklon tangente v tej točki



$$f(x+\delta x) \approx f(x) + \frac{df}{dx}\cdot\delta x$$

bližu točke x

približek

f' , $\dot{f}$, $\frac{dy}{dx}$ ... in ulomek!

Def: $\frac{df}{dx} = \lim_{dx \to 0} \frac{f(x+dx) - f(x)}{dx}$

uporaba: 1. zveze med fizikalnimi količinami (hitrost, moč)

2. ekstrem funkcij

Splošne zveze:

1. vsota $(f(x) + g(x))' = f'(x) + g'(x)$

2. produkt $(f(x) \cdot g(x))' = f'(x) \cdot g(x) + f(x) g'(x)$

3. količnik $\left(\frac{f(x)}{g(x)}\right)' = \frac{f'(x) g(x) - f(x) g'(x)}{g^2(x)}$

4. KOMPOZITUM $\sin(x^2)$

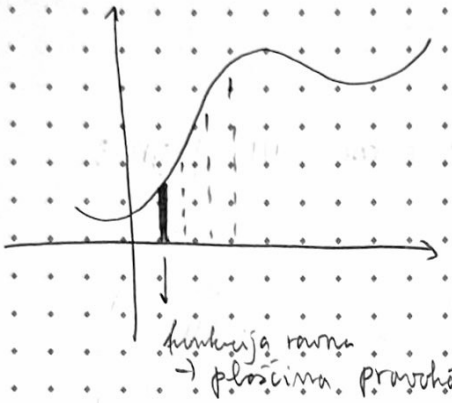
$(f(g(x)))'$

$= f'(g(x)) \cdot g'(x)$

$\int f(x)\,dx = F(x)$	$f(x) = F'(x)$
C	0
x^n	$n x^{n-1}$
e^x	e^x
e^{ax}	$a e^{ax}$
a^x	$\ln(a)\, a^x$
$\log x$	$\frac{1}{x}$

$\sin x$	$\cos x$
$\cos x$	$-\sin x$

4. INTEGRALI ... vsota neskončno majhnih kosčkov



ploščina ravninskega lika → ploščina pravokotnika

$$F(x) = \int f(x)\,dx + C$$

NEDOLOČEN INTEGRAL

$$F = \int_a^b f(x)\,dx$$

$$= F(b) - F(a)$$

DOLOČENI INTEGRAL

zamenjava spremenljivke

$$I = \int f(ax)\,dx \quad , \text{ nova spr. } z = ax$$

$$dz = a\,dx \qquad dx = \frac{dz}{a}$$

$$I = \int f(z)\,\frac{dz}{a}$$

$f(x)$	$F(x) = \int f(x)\,dx$
C	Cx
x^n	$\frac{1}{n+1}\,x^{n+1}$
e^x	e^x
e^{ax}	$\frac{1}{a}\,e^{ax}$

a^x	$\frac{a^x}{\ln a}$
$\frac{1}{x}$	$\ln x$
$\sin x$	$-\cos x$
$\cos x$	$\sin x$

7 KORAKOV ZA REŠEVANJE NALOG

1. Preberi + prepoznaj fizikalne teme in pojme
2. Predstava: mentalni film
3. film pretvori v skico → zakoni
4. iz zakonov napišemo enačbo
5. ponavljate 4., dokler nimam dovolj enačb
6. Rečim enačbe (matematika)
7. Interpretacija (ali se sklada s filmom?)

NALOGE:

1.) a) $a = \frac{dv}{dt} = \cancel{\lim_{\Delta t \to 0}} \frac{v(t+dt) - v(t)}{dt}$

ker je $v(t)$ premica

$t = 1\,s : v(1\,s) = 8\,m/s$

$t + dt = 2\,s : v(2\,s) = 6\,m/s$

$\frac{6\,m/s - 8\,m/s}{2\,s - 1\,s}$

$= \underline{\underline{-2\,m/s^2}}$

ODGOVOR: $\underline{\underline{2\,m/s^2}}$ (ker je pojemek)

b) $v(t) = \frac{ds(t)}{dt}$

$\Rightarrow \int ds(t) = \int v(t) \cdot dt$

$\int x^n dx = \frac{1}{n+1}$

$v(t) = v_0 + at$

$f(x) = (n + kx)$

$s(t) = \int (v_0 + at)\, dt$

$= \int v_0\, dt + \int at\, dt$

$= \underline{v_0 \cdot t + a \frac{t^2}{2}}$

$v_0 = 10\ m/s$

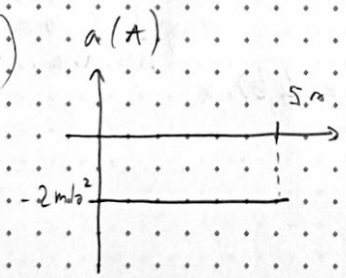
$a = -2\ m/s^2$

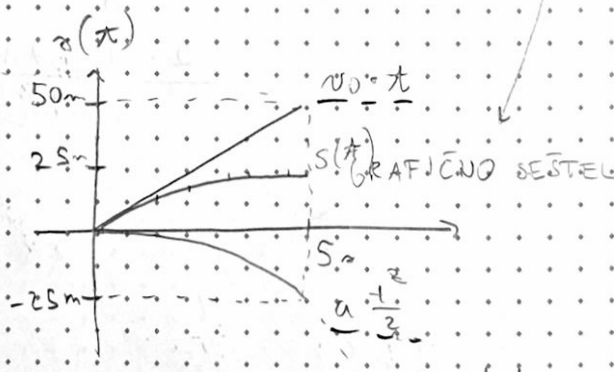
$s(t_{ustav}) = s(t = 5s)$

$= 10\ m/s \cdot 5s + \frac{1}{2}(-2\ m/s^2)(5s)^2$

$= 50m - 25m = \underline{25m}$

c)





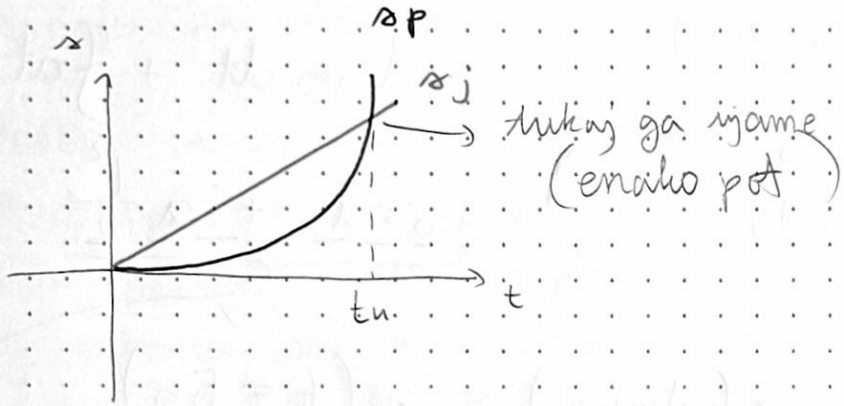
2)

a) kdaj ga ujame: ko $s_j(t_u) = s_p(t_u)$

$s_j(t) = v_0 \cdot t$

$v_0 = 60\ km/h = 16{,}67\ m/s$

$s_p(t) = \frac{1}{2} a t^2$



$s_j(t_u) = s_p(t_u)$

$v_0 \cdot t_u = \frac{1}{2} a t_u^2$ → s tem smo izločili enačbo $t_u = 0$

$t_u = \frac{2 v_0}{a}$ (Odkod sta se ze vzela ampak to nas ne zanima.)

$t_u = \frac{2 \cdot 16{,}67\ m/s}{5\ m/s^2} = 6{,}67\ s$

b)

$\Delta x \to 0$ Δx

$3t - 2$ 2. $x = -4t^2 - 2$ 3. $x = \frac{2}{t^2}$

3

$v = -8t$

$v = \frac{(2)(2t) - (0)(t^4)}{t^4}$

$v = \frac{4t}{t^4} = \frac{4}{t^3}$

$x = -2$

$v = 0$

or $x = 2t^{-2}$

$v = -2(2(t^{-3}))$

$= \frac{-4}{t^{-3}}$

$\frac{dv}{dt} = \frac{d^2x}{dt^2}$

$4 - 27t + t^3$

$= \frac{dx}{dt} = \frac{-27 + 3t^2}{2}$

$) = \frac{dv}{dt} = \frac{d^2x}{dt^2} = 6t$

)

$) = v_0 - kt^2$

$0,1\ m/s^3$

$v(t) = \frac{ds}{dt}$

$ds = v(t) \cdot dt \ / \cdot \int$

$s = \int v(t) \cdot dt$

$s = \int (v_0 - kt^2)\, dt$

$s = v_0 \cdot t - k \cdot \frac{t^3}{3}$ |

$$a_p = \frac{dv}{dt}$$

$$\Rightarrow dv = a_p \cdot dt \quad / \int$$

$$v = \int_0^t a_p \cdot dt$$

$$v_p = a_p \cdot t$$

$$v_p = \frac{ds_p}{dt}$$

$$ds_p = v_p \, dt$$

$$s_p = \int v_p \, dt = \int a_p \cdot t \, dt$$

$$s_p = \frac{a_p t^2}{2}$$

$$s_1 = s_p$$

$$v_0 \cdot t - k \cdot \frac{t^3}{3} = \frac{a_p t^2}{2}$$

$$t\left(v - \frac{k t^2}{3}\right) = \frac{a_p t^2}{2}$$

$$v - \frac{k t^2}{3} - \frac{a_p t}{2} = 0 \quad / \cdot 6$$

$$6v - 2kt^2 - 3a_p t = 0$$

$$6v = 2kt^2 + 3a_p t$$

$$t(2kt + 3a_p) = 6v$$